Muhammad Alaric Athaya Iskandar
25/555070/PA/23257
CSA

Assignment 2 Report Image Enhancement

I implemented image enhancement techniques in this assignment. The algorithms I created were able to fix dark, overexposed, low - contrast, and blurry images. I used an image that can be seen as dark which shows a man wearing a black hoodie with treelines and water in the background. I chose this to make sure that the algorithm can enhance the image and expose more of the background as well.

Methods:

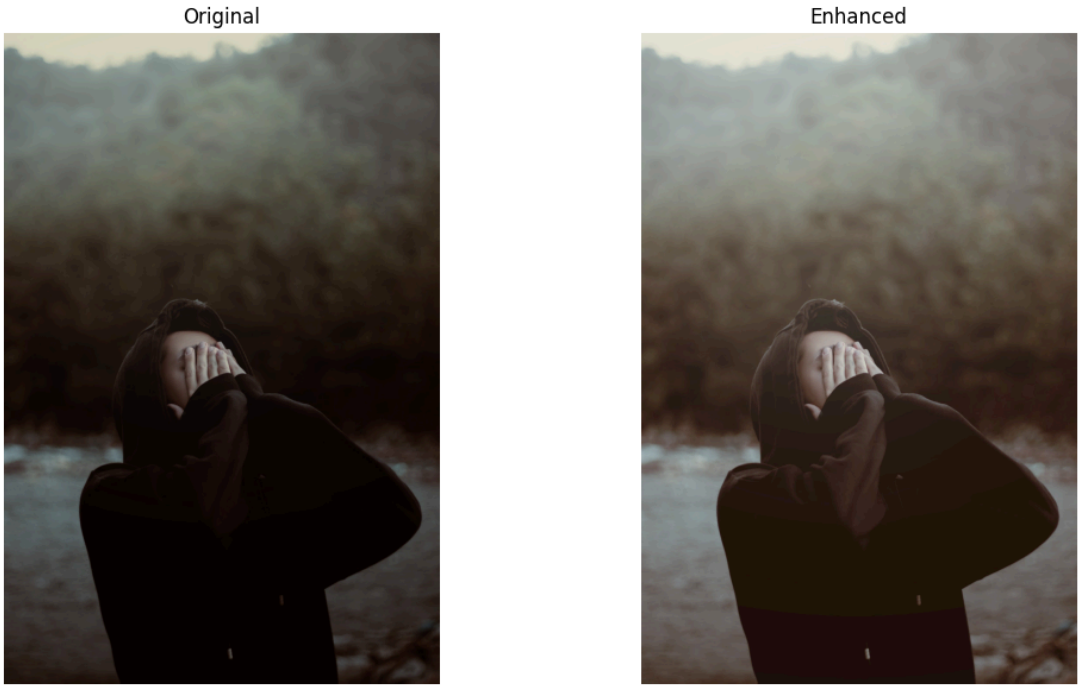
- Power - law (gamma) transformation (s = c.r^gamma) = this fixes dark where gamma < 1, or overexposed gamma > 1 images.
- Histogram equalization — fixes low contrast.
- Laplacian sharpening (g = f - c.Laplacian(f)) — fixes blur.
- Denoising (median / non-local means) — applied before brightening dark images to avoid amplifying sensor noise.

Results:

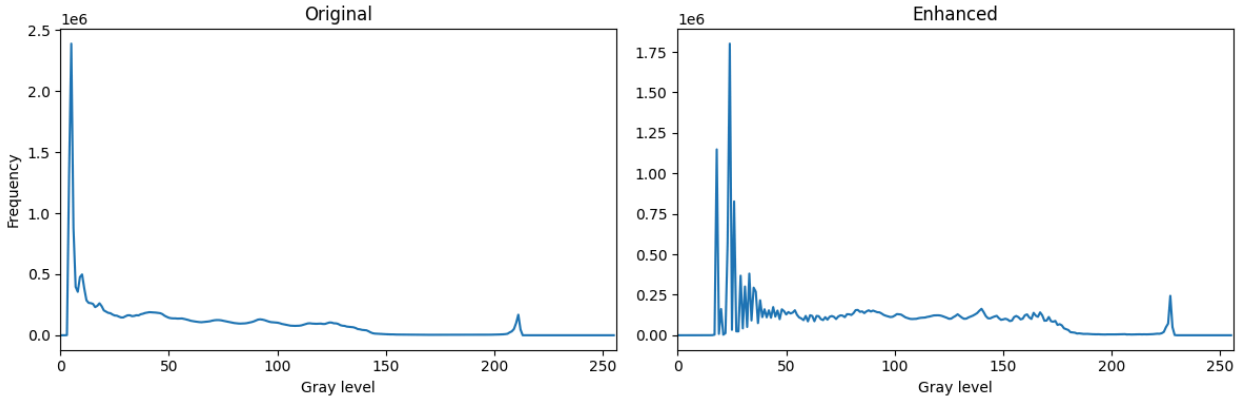
In this photo, the background such as the sky, treeline and water can be seen getting brighter and showing more texture that was black before. The water and foliage can be seen more. There is no grain or noise that appeared because this photo wasn't that intense to trigger the denoising code. The person in the photo still stays dark since it is a blacklit silhouette which is darker than the background in the first place, the gamma correction fixes everything proportionally instead of just doing the same towards subject and background.

- Brightness = 52.0<90: This confirms that the gamma correction (and denoising) was executed due to the fact that the image is clearly underexposed by the pipeline's threshold. After that, the brightness rose which can be seen by the human eye.
- blur_score = 7.8 < 100: This triggered the blur/sharpening branch. Even though the image looks already sharp, it is still measured by blur_score which is under the 100 threshold, so it caused the Laplacian sharpening to run. The enhanced blur_score can

be seen that it went down slightly. This is caused by the moderate blur_score ran together with heavy brightening/denoising which leaves an image measuring blurrier even though it still looks fine to the human eye since denoising softens fine texture.



Histogram:



The histogram shows how many pixels exist at each gray level from 0 to 255. The original histogram showed that almost all pixels are crowded near the level of 0 to 10 with a small and long part spiking to 250, and a small one around 210. This shows a dark image.

The enhanced version,there is still a same spike from around 10 to 20-25. The values are more spread out across the low - to - mid range instead of being crammed at the bottomline. This proves that gamma correction worked as it doesn't flatten the histogram. All it does is itjust moves and pulls the values upward, but keeping the same shape.